

Parallel & Concurrent Programming

kotlinlang.org/education

Definition

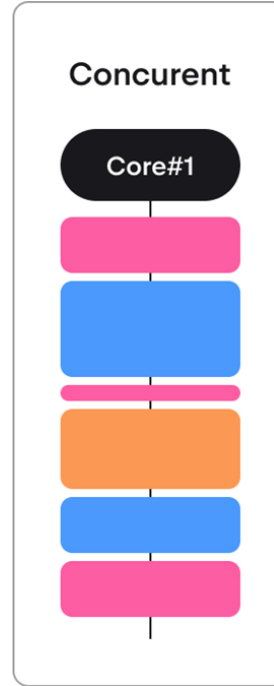
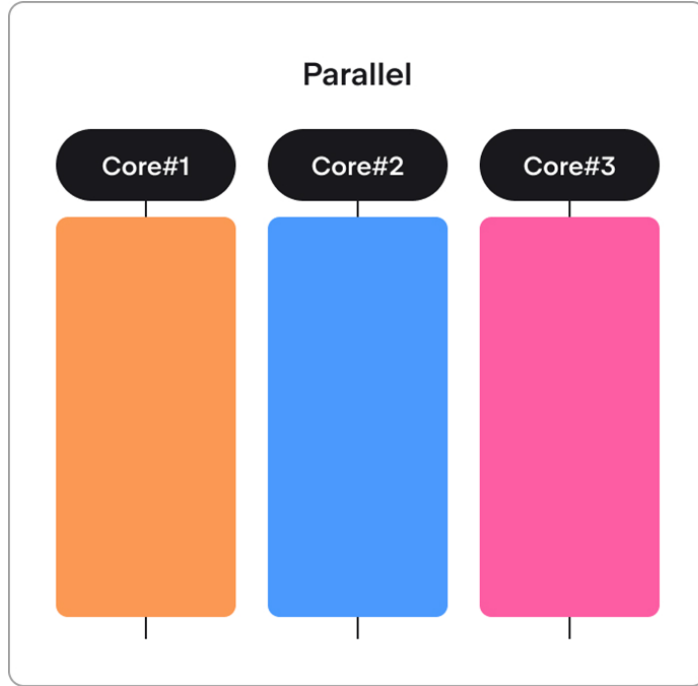
According to Wikipedia:

- **Parallel** computing is a type of computing “in which many calculations or processes are carried out **simultaneously**”.
- **Concurrent** computing is a form of computing in which several computations are executed **concurrently** – in overlapping time periods – instead of sequentially.
- It is possible to have parallelism without concurrency, and concurrency without parallelism.

Motivation

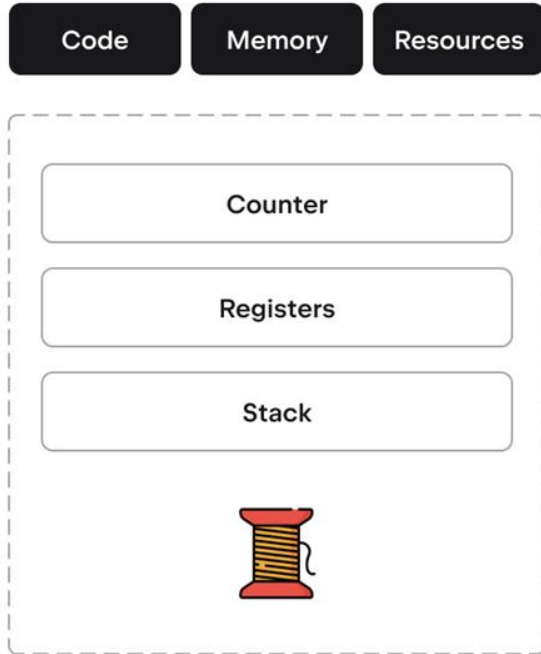
- Faster runtime
- Improved responsiveness

Parallelism vs concurrency

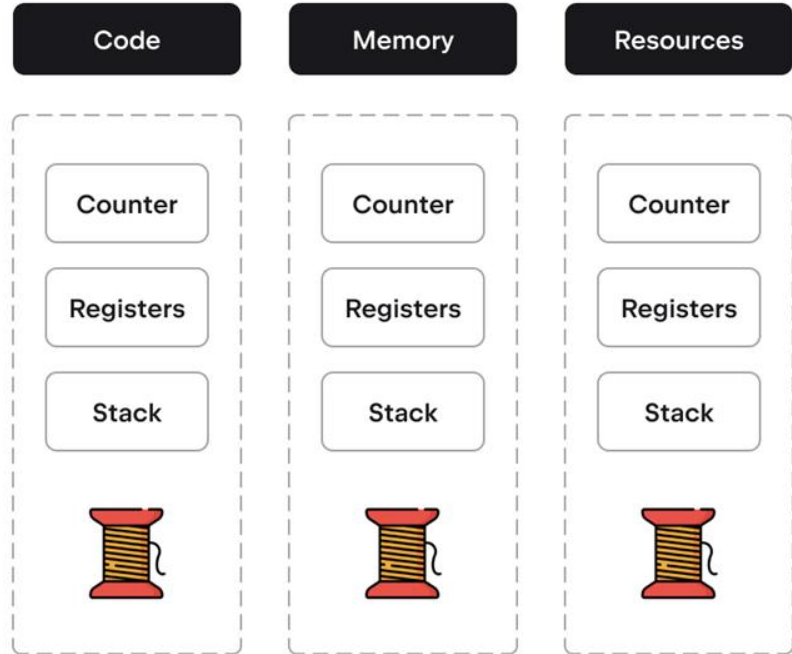


Concurrency: processes vs threads

Single-threaded process

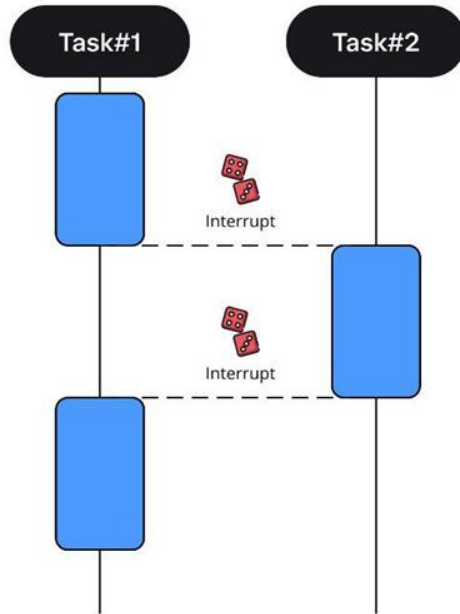


Multi-threaded process

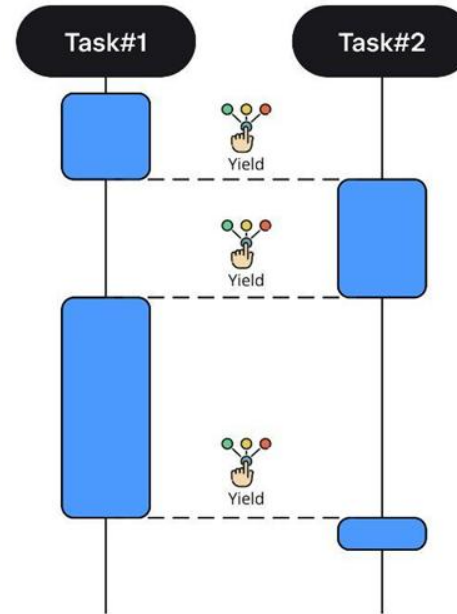


Preemptive vs cooperative scheduling

Preemptive: OS interrupts tasks



Cooperative: task yields control



Parallel and concurrent Programming in the JVM

- The JVM has its own scheduler
 - It is independent from the OS scheduler
 - A JVM thread != an OS thread
 - => Multithreaded JVM apps can run on a single-threaded OS
- (DOS) JVM threads are either daemons or user threads.
- The app stops when all **user threads** are done.
- The JVM does not wait for daemon threads to finish.

Parallel programming in the JVM

2 Java packages

- `java.lang` contains basic primitives: `Runnable`, `Thread`, etc
- `java.util.concurrent` contains synchronization primitives and concurrent data structures

Kotlin package

- `kotlin.concurrent` — Wrappers and extensions for Java classes

Throwback: Single abstract method interfaces

```
@FunctionalInterface
public interface Runnable {
    public abstract void run();
}
```

Interface with a single method. We can instantiate it with a lambda.

```
class RunnableWrapper(val runnable: Runnable)

val myWrapperObject =
    RunnableWrapper(
        object : Runnable {
            override fun run() {
                println("I run")
            }
        }
    )
val myWrapperLambda = RunnableWrapper { println("yo") }
```


Ways to create threads

You can inherit from the Thread class, which also implements Runnable.

```
class MyThread : Thread() {  
    override fun run() {  
        println("${currentThread()} is running")  
    }  
}  
  
fun main() {  
    val myThread = MyThread()  
    myThread.start()  
}
```

run vs start

Never call Thread.run()!

run will execute on *your* thread, while start will create a new thread where run will be executed.

```
fun main() {  
    val myThread1 = MyThread()  
    myThread1.start() // OK  
    val myThread2 = MyThread()  
    myThread2.run() // Current thread gets blocked  
}
```

Ways to create threads

You can implement the `Runnable` interface and pass it to a thread. You can pass the same `Runnable` to several threads.

```
fun main() {  
    val myRunnable = Runnable { println("Sorry, gotta run!")  
}  
    val thread1 = Thread(myRunnable)  
    thread1.start()  
    val thread2 = Thread(myRunnable)  
    thread2.start()  
}
```

Ways to create threads

Kotlin has an even simpler way to create threads, but under the hood the same old thread is created and started.

```
import kotlin.concurrent.thread

fun main() {
    val kotlinThread = thread {
        println("I start instantly, but you can pass an option to start me later")
    }
}
```

This is the preferable way to create threads.

Thread properties

A thread's properties cannot be changed after it is started.

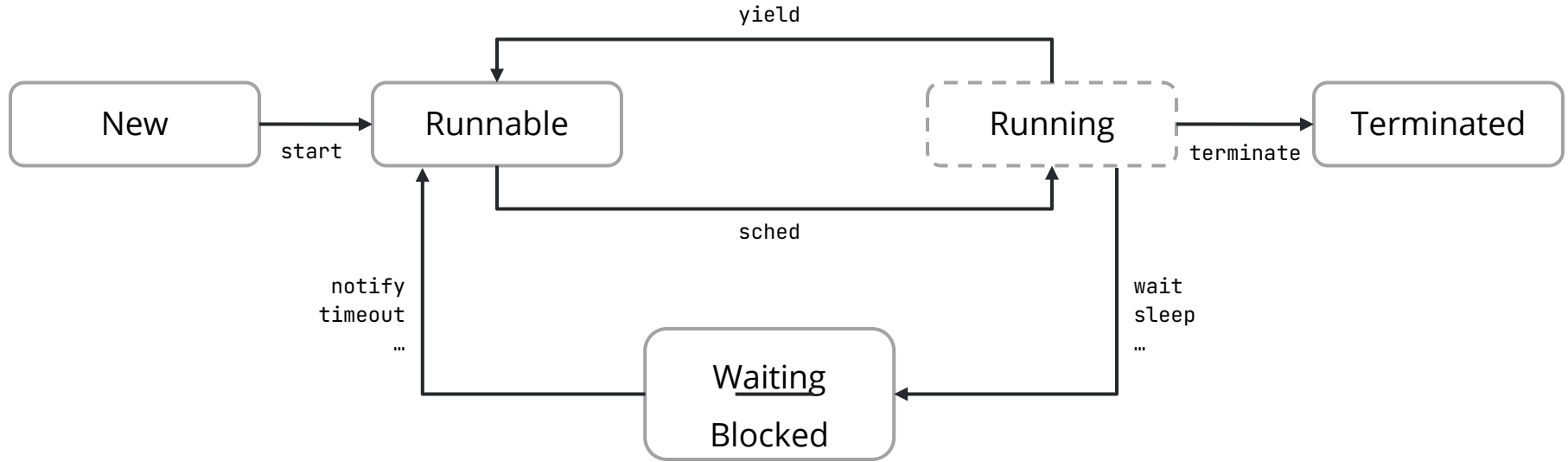
Main properties of a thread:

- `id: Long` — This is the thread's identifier
- `name: String`
- `priority: Int` — This can range from 1 to 10, with a larger value indicating higher priority
- `daemon: Boolean`
- `state: Thread.State`
- `isAlive: Boolean`

State of a thread

state	isAlive
NEW	false
RUNNABLE	true
BLOCKED	true
WAITING	true
TIMED_WAITING	true
TERMINATED	false

State of a thread



Ways to manipulate a thread's state

- `val myThread = thread { ... }` — Creates a new thread
- `myThread.start()` — Starts a thread
- `myThread.join()` — Causes the current thread to wait for another thread to finish
- `sleep(...)` — Puts the current thread to sleep
- `yield()` — *Tries* to step back
- `myThread.interrupt()` — *Tries* to interrupt a thread
- `myThread.isInterrupted()` — Checks whether thread was interrupted
- `interrupted()` — Checks and clears the interruption flag

sleep, join, yield, interrupt

- The `sleep` and `yield` methods are only applicable to the current thread, which means that you cannot suspend another thread.
- All blocking and waiting methods can throw `InterruptedException`

Classic worker

```
class ClassicWorker : Runnable {  
    override fun run() {  
        try {  
            while (!Thread.interrupted()) {  
                // do stuff  
            }  
        } catch (e: InterruptedException) {} // absolutely legal empty catch block  
    }  
}
```

Parallelism and shared memory: Examples of problematic interleaving

Parallel threads have access to the same shared memory.

This often leads to problems that cannot arise in a single-threaded environment.

```
class Counter {  
    private var c = 0  
  
    fun increment() {  
        c++  
    }  
    fun decrement() {  
        c--  
    }  
    fun value(): Int {  
        return c  
    }  
}
```

Both operations on c are single, simple statements.

However, even simple statements can be translated into multiple steps by the virtual machine, and those steps can be interleaved.

Parallelism and shared memory: Examples of problematic interleaving

Parallel threads have access to the same shared memory.

This often leads to problems that cannot arise in a single-threaded environment.

```
class Counter {  
    private var c = 0  
  
    fun increment() {  
        c++  
    }  
    fun decrement() {  
        c--  
    }  
    fun value(): Int {  
        return c  
    }  
}
```

Suppose both Thread#1 and Thread#2 invoke increment at the same time. If the initial value of c is 0, their interleaved actions might follow this sequence:

- T#1: Read value 0 from c.
- T#2: Read value 0 from c.
- T#1: Increment value — result is 1.
- T#1: Write result 1 to c.
- T#2: Increment value — result is 1.
- T#2: Write result 1 to c.

Synchronization mechanisms

- Mutual exclusion, such as Lock and the synchronized keyword
- Concurrent data structures and synchronization primitives
- Atomics, which work directly with shared memory (DANGER ZONE)

Locks

```
class LockedCounter {  
    private var c = 0  
  
    private val lock = ReentrantLock()  
  
    fun increment() {  
        lock.withLock { c++ }  
    }  
  
    // same for other methods  
    ...  
}
```

The lock interface

- `lock.lock()` — Acquires the lock
- `lock.tryLock()` — Tries to acquire the lock
- `lock.unlock()` — Releases the lock
- `lock.withLock { }` — Executes a lambda with the lock held (has try/catch inside)
- `lock.newCondition()` — Creates a *condition variable* associated with the lock

Conditions

```
class PositiveLockedCounter {  
    private var c = 0  
    private val lock = ReentrantLock()  
    private val condition = lock.newCondition()  
  
    fun increment() {  
        lock.withLock {  
            c++  
            condition.signal()  
        }  
    }  
  
    fun decrement() {  
        lock.withLock {  
            while (c == 0) {  
                condition.await()  
            }  
            c--  
        }  
    }  
  
    fun value(): Int {  
        return lock.withLock { c }  
    }  
}
```

A condition allows a thread holding a lock to *wait* until another thread *signals* it about a certain event. Internally, the `await` method releases the associated lock upon call, and acquires it back before finally returning it again.

The ReentrantLock class

- `ReentrantLock` – Allows the lock to be acquired multiple times by the same thread
- `lock.getHoldCount()` – Gets the number of holds on this lock by the current thread
- `lock.queuedThreads()` – Gets a collection of the threads waiting on this lock
- `lock.isFair()` – Checks the *fairness* of the lock

The synchronized statement

In the JVM, every object has an *intrinsic* lock associated with it (aka a *monitor*).

```
class Counter {  
    private var c = 0  
  
    fun increment() {  
        synchronized(this) { c++ }  
    }  
  
    ...  
}
```

Synchronized method

Java

```
public class SynchronizedCounter {  
    private int c = 0;  
  
    public synchronized void increment() {  
        c++;  
    }  
  
    ...  
}
```

Kotlin

```
class SynchronizedCounter {  
    private var c = 0  
  
    @Synchronized  
    fun increment() {  
        c++  
    }  
  
    ...  
}
```

The ReadWriteLock class

ReadWriteLock allows multiple readers to access a resource concurrently but only lets a single writer modify it.

- `rwLock.readLock()` – Returns the read lock
- `rwLock.writeLock()` – Returns the write lock
- `rwLock.read { ... }` – Executes lambda under a read lock
- `rwLock.write { ... }` – Executes lambda under a write lock

The ReadWriteLock Class

```
class PositiveLockedCounter {  
    private var c = 0  
    private val rwLock = ReadWriteReentrantLock()  
  
    fun increment() {  
        rwLock.write { c++ }  
    }  
  
    fun decrement() {  
        rwLock.write { c-- }  
    }  
  
    fun value(): Int {  
        return rwLock.read { c }  
    }  
}
```

Concurrent blocking collections

`java.util.concurrent` is a Java package that implements both blocking and non-blocking concurrent collections, such as:

- `SynchronousQueue` – One-element rendezvous channel
- `ArrayBlockingQueue` – Fixed-capacity queue
- `LinkedBlockingQueue` – Unbounded blocking queue
- `PriorityBlockingQueue` – Unbounded blocking priority queue

Concurrent non-blocking collections

`java.util.concurrent` is a Java package that implements both blocking and non-blocking concurrent collections, such as:

- `ConcurrentLinkedQueue` – Non-blocking unbounded queue
- `ConcurrentLinkedDeque` – Non-blocking unbounded deque
- `ConcurrentHashMap` – Concurrent unordered hash-map
- `ConcurrentSkipListMap` – Concurrent sorted hash-map

Synchronization primitives

`java.util.concurrent` also implements concurrent data structures and synchronization primitives.

- Exchanger – Blocking exchange
- Phaser – Barrier synchronization

Java Memory Model: Weak behaviors

There are no guarantees when it comes to ordering!

Possible outputs:

```
class OrderingTest {  
    var x = 0  
    var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```

- 0, 0
- 0, 1
- 1, 1
- 1, 0

Java Memory Model: Weak behaviors

There are no guarantees when it comes to progress!

```
class ProgressTest {  
    var flag = false  
    fun test() {  
        thread {  
            while (!flag) {}  
            println("I am free!")  
        }  
        thread { flag = true }  
    }  
}
```

Possible outputs:

- "I am free!"
- ...
- ...
- ...
- hang!

Java Memory Model: Weak behaviors

There are no guarantees when it comes to progress!

```
class ProgressTest {  
    var flag = false  
    fun test() {  
        thread {  
            while (true) {}  
            println("I am free!")  
        }  
        thread { flag = true }  
    }  
}
```

Possible outputs:

- "I am free!"
- ...
- ...
- ...
- hang!

JMM: Data-Race-Freedom Guarantee

But what does JMM guarantee?

Well-synchronized programs have **simple interleaving semantics**.

JMM: Data-Race-Freedom Guarantee

But what does JMM guarantee?

Well-synchronized programs have **simple interleaving semantics**.

Well-synchronized = Data-race-free

Simple interleaving semantics = Sequentially consistent semantics

Data-race-free programs have **sequentially consistent** semantics

JMM: Volatile fields

Volatile fields can be used to restore **sequential consistency**.

```
class OrderingTest {  
    @Volatile var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```

```
class ProgressTest {  
    @Volatile var flag = false  
    fun test() {  
        thread {  
            while (!flag) {}  
            println("I am free!")  
        }  
        thread { flag = true }  
    }  
}
```

JMM: Volatile fields

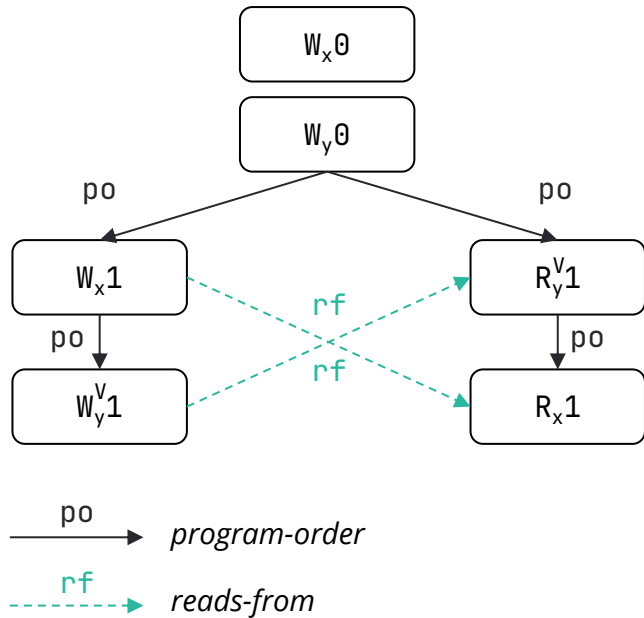
Volatile variables can be used for synchronization.

```
class OrderingTest {  
    var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```

How do we know there is enough synchronization?

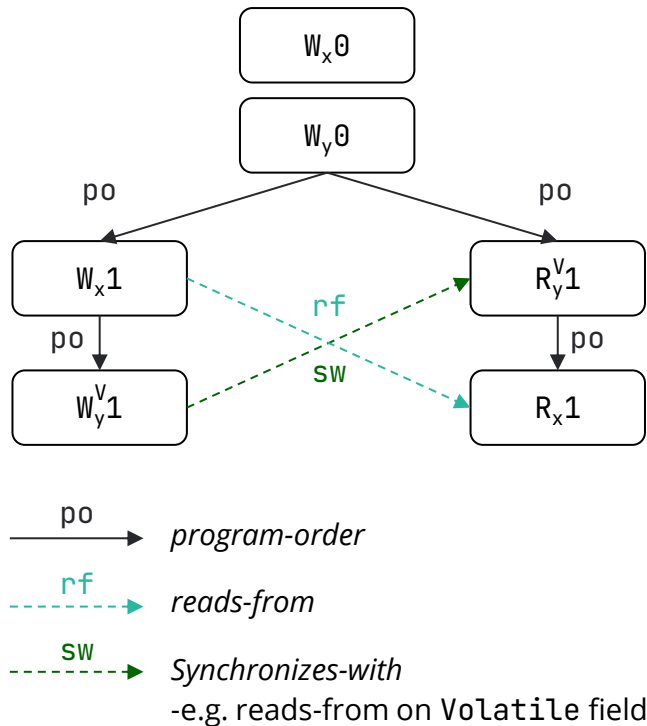
JMM: Happens-before relation

```
class OrderingTest {  
    var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```



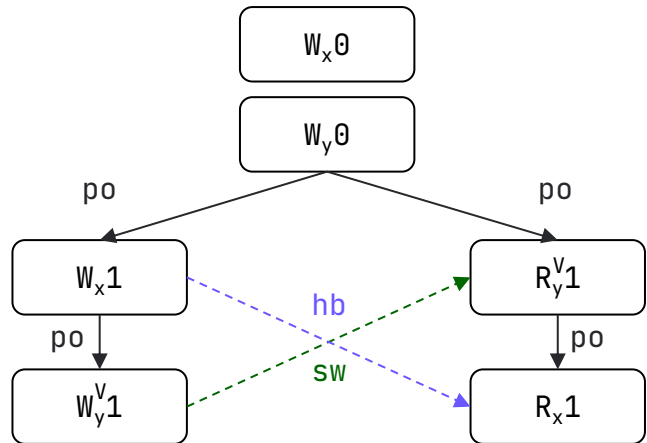
JMM: Happens-before relation

```
class OrderingTest {  
    var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```



JMM: Happens-before relation

```
class OrderingTest {  
    var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```



→ po → *program-order*

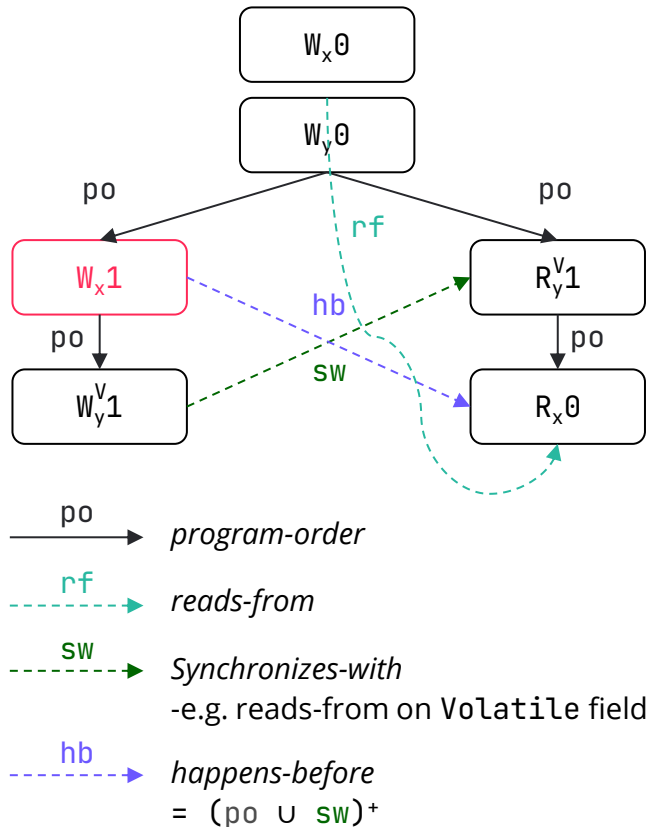
-.-> rf -.-> *reads-from*

-.-> sw -.-> *Synchronizes-with*
-e.g. reads-from on Volatile field

-.-> hb -.-> *happens-before*
= (po ∪ sw)⁺

JMM: Happens-before relation

```
class OrderingTest {  
    var x = 0  
    @Volatile var y = 0  
    fun test() {  
        thread {  
            x = 1  
            y = 1  
        }  
        thread {  
            val a = y  
            val b = x  
            println("$a, $b")  
        }  
    }  
}
```



JMM: Synchronizing actions

- Read and write for volatile fields
- Lock and unlock
- Thread run and start, as well as finish and join

JMM: DRF-SC again

Two events form a data race if:

- Both are memory accesses to the same field.
- Both are **plain** (non-atomic) accesses.
- At least one of them is a `write` event.
- They are not related by *happens before*.

Data-race-free programs have **sequentially consistent** semantics

A program is data-race-free if, for every possible execution of this program, no two events form a data race.

JMM: Atomics

But what about atomic operators on shared variables?

```
class Counter {  
    private val c = AtomicInteger()  
  
    fun increment() {  
        c.incrementAndGet()  
    }  
  
    fun decrement() {  
        c.decrementAndGet()  
    }  
  
    fun value(): Int {  
        return c.get()  
    }  
}
```

JMM: Atomics

Atomic classes from package the `java.util.concurrent.atomic` package:

- `AtomicInteger`
- `AtomicLong`
- `AtomicBoolean`
- `AtomicReference`

And their array counterparts:

- `AtomicIntegerArray`
- `AtomicLongArray`
- `AtomicReferenceArray`

JMM: Atomics

- `get()` – Reads a value with volatile semantics
- `set(v)` – Writes a value with volatile semantics
- `getAndSet(v)` – Atomically exchanges a value
- `compareAndSet(e, v)` – Atomically compares a value of atomic variable with the expected value, `e`, and if they are equal, replaces content of atomic variable with the desired value, `v`; returns a boolean indicating success or failure.
- `compareAndExchange(e, v)` – Atomically compares a value with an expected value, `e`, and if they are equal, replaces with the desired value, `v`; returns a read value.
- `getAndIncrement()`, `addAndGet(d)`, etc – Perform Atomic arithmetic operations for • numeric atomics (`AtomicInteger`, `AtomicLong`).
- ...

JMM: Atomics

Methods of atomic classes:

- ...
- `getXXX()`
- `setXXX(v)`
- `weakCompareAndSetXXX(e, v)`
- `compareAndExchangeXXX(e, v)`

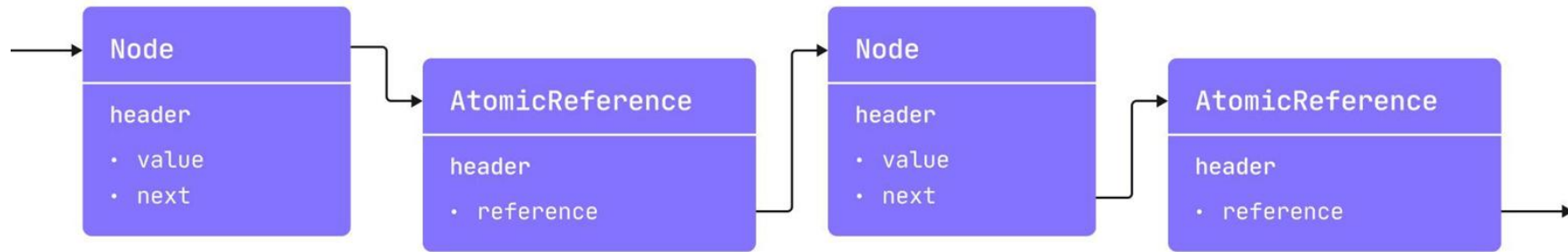
In these cases, XXX is an access mode: Acquire, Release, Opaque, Plain

You can learn more about Java Access Modes here:

<https://gee.cs.oswego.edu/dl/html/j9mm.html>

JMM: Atomics Problem

```
class Node<T>(val value: T) {  
    val next = AtomicReference<Node<T>>()  
}
```



JMM: Atomic field updaters

Use AtomicXXXFieldUpdater classes to directly modify volatile fields:

```
class Counter {  
    @Volatile private var c = 0  
    companion object {  
        private val updater = AtomicIntegerFieldUpdater.newUpdater(Counter::class.java, "c")  
    }  
    fun increment() {  
        updater.incrementAndGet(this)  
    }  
    fun decrement() {  
        updater.decrementAndGet(this)  
    }  
    fun value(): Int {  
        return updater.get(this)  
    }  
}
```

Starting from JDK9, there is also the VarHandle class, which serves a similar purpose.

Kotlin: AtomicFU

The AtomicFU library is a recommended way to use atomic operations in Kotlin:

<https://github.com/Kotlin/kotlinx-atomicfu>

```
class Counter {  
    private val c = atomic(0)  
    fun increment() {  
        c += 1  
    }  
    fun decrement() {  
        c -= 1  
    }  
    fun value(): Int {  
        return c.value  
    }  
}
```

- It provides AtomicXXX classes with API similar to Java atomics.
- Under the hood *compiler plugin* replaces usage of atomics to AtomicXXXFieldUpdater or VarHandle.
- It also provides convenient extension functions, e.g.
c.update { it + 1 }

Thank you

